

Bayesian Inference

Advanced Statistics Lab

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Bayes Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

In statistical modelling:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

Bayesian vs Frequentist

This is in contrast with a frequentist (maximum likelihood) approach, in which we aim to estimate non-random, fixed parameters using the likelihood of the data given those parameters only ($p(y|\theta)$).

The overall likelihood of the data ($p(y)$) only serves as a scaling constant:

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

Estimating A Normal Mean

In a Bayesian framework, we formulate our problem in terms of the assumed distribution of our data (likelihood) and the prior distributions of our parameters we wish to estimate.

$$\mathbf{y} \sim \mathbf{N}(\mu, \sigma_e^2)$$

$$\mu \sim N(\mu_{prior}, \sigma_{prior}^2)$$

brms Mean Example

```
library(brms)
library(ggplot2)
library(knitr)
library(tidyverse)

data(iris)

# this gives us the default brms priors
get_prior(Sepal.Width ~ 1, data = iris)

# set a custom prior for the intercept
priors = set_prior("normal(0,0.2)", class = "Intercept")
```

Fit The Model

```
fit <- brm(Sepal.Width ~ 1,  
  data = iris,  
  family = gaussian(),  
  prior = priors,  
  sample_prior = TRUE,  
  silent = 2,  
  refresh = 0  
)
```

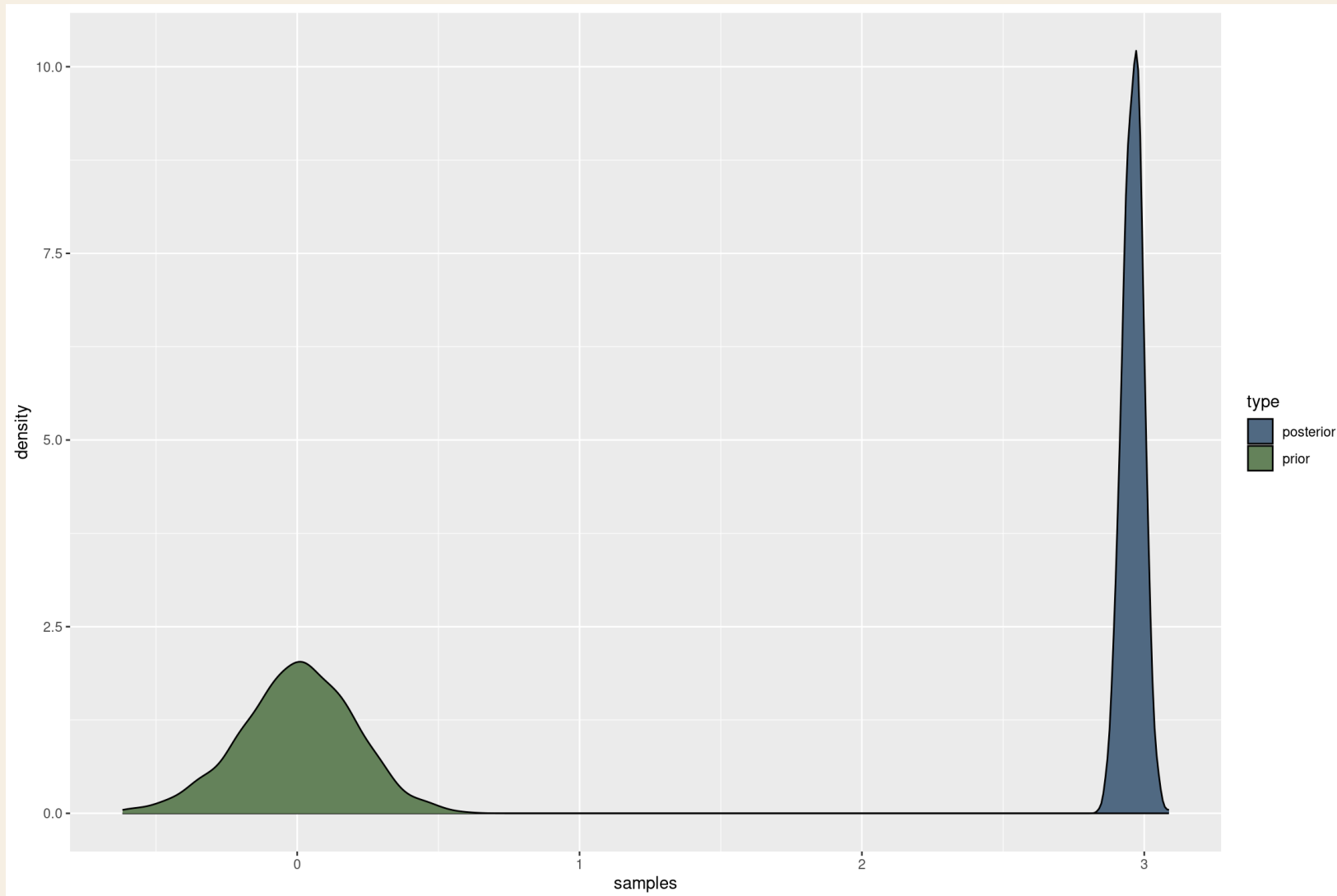
Prior And Posterior Samples

```
draws = as_draws_array(fit)[,1,, drop = TRUE]

samples = rbind(
  data.frame(type = "prior",
             samples = as.numeric(draws[, "prior_Intercept"])),
  data.frame(type = "posterior",
             samples = as.numeric(draws[, "b_Intercept"]))
)

kable(
  samples %>% group_by(type) %>% summarize(
    n = n(),
    mean = mean(samples),
    sd = sd(samples)
  ),
  digits = 3
)
```

Prior And Posterior



Updating Priors From Existing Evidence

Bayesian Inference allows us to incorporate actual prior evidence in the estimation of parameters.

We might have conducted a previous experiment and arrived at an estimate of the mean of our observations.

Then we do another experiment of similar setup and want to update our mean estimate.

Prior Updating

$$p(\theta|y)_{FirstExperiment} \propto p(y|\theta)p(\theta)$$

Plugging in the posterior from the first experiment as the prior for the second:

$$p(\theta|y)_{SecondExperiment} \propto p(y|\theta)p(\theta|y)_{FirstExperiment}$$

Update Prior In R

```
iris_first = iris[1:50,]

priors = set_prior("normal(0,4)", class = "Intercept")

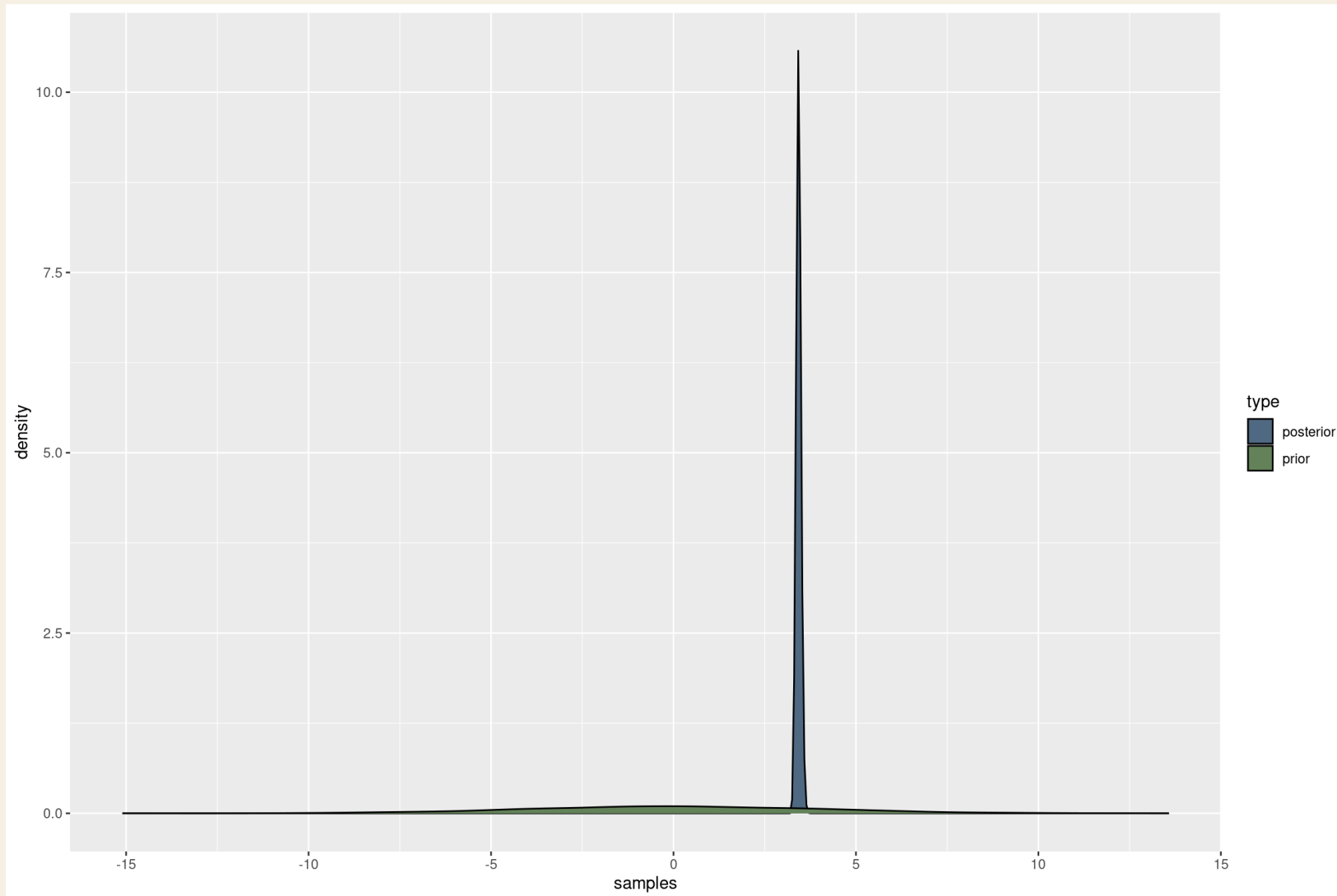
fit_first <- brm(Sepal.Width ~ 1,
  data = iris_first,
  family = gaussian(),
  prior = priors,
  sample_prior = TRUE,
  silent = 2,
  refresh = 0
)

draws_first = as_draws_array(fit_first)[,1,, drop = TRUE]
```

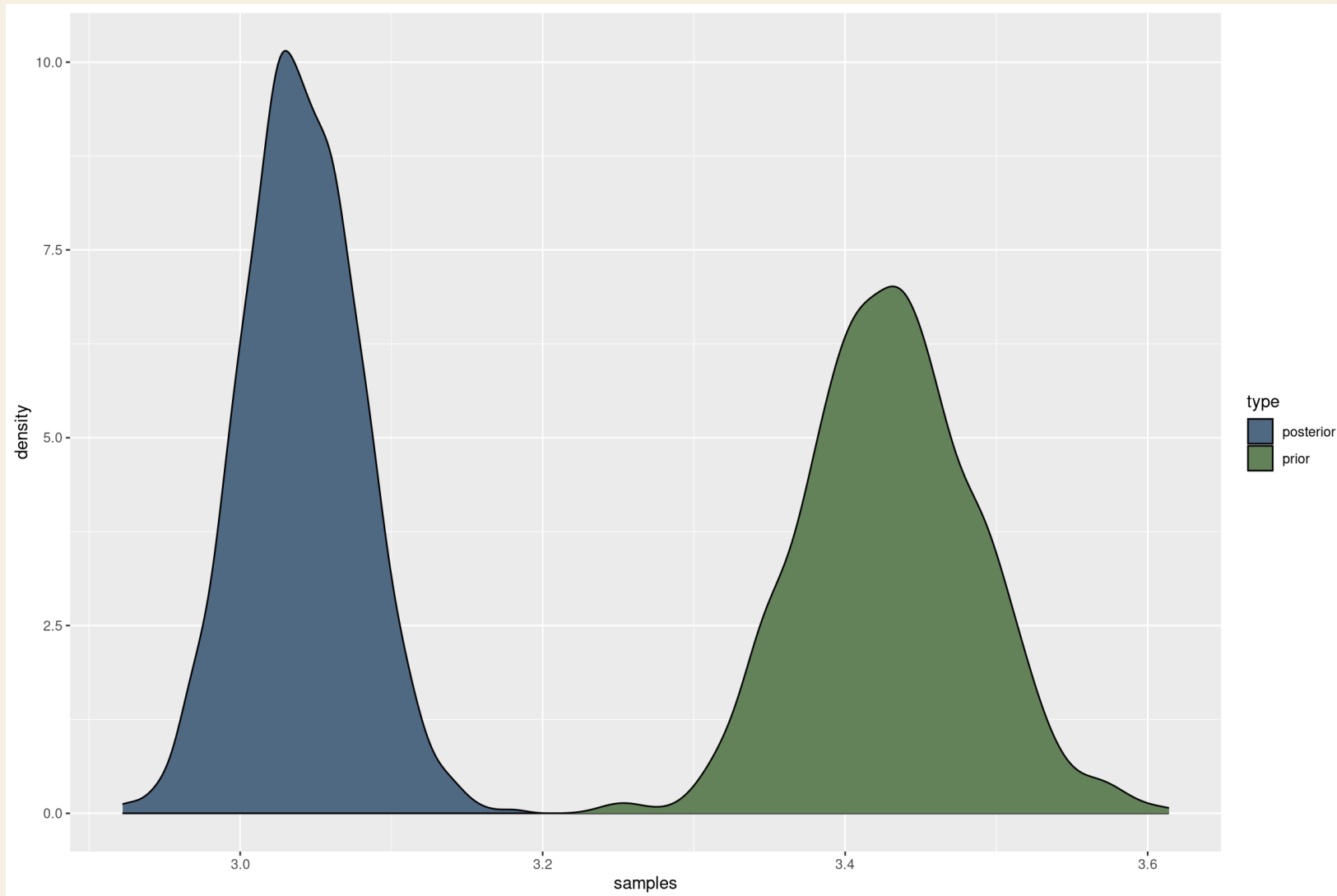
Use Posterior As Prior

```
samples_first = rbind(  
  data.frame(type = "prior",  
             samples = as.numeric(draws_first[, "prior_Intercept"])),  
  data.frame(type = "posterior",  
             samples = as.numeric(draws_first[, "b_Intercept"]))  
)  
  
mu_prior = mean(samples_first$samples[samples_first$type == "posterior"])  
sd_prior = sd(samples_first$samples[samples_first$type == "posterior"])  
  
priors = set_prior(  
  paste("normal(", mu_prior, ",", sd_prior, ")", sep = ""),  
  class = "Intercept"  
)
```

First Experiment



Second Experiment



Multivariate Animal Model In brms

$$\mathbf{Y} = \mathbf{XB} + \mathbf{ZU} + \mathbf{E}$$

Likelihood:

$$\mathbf{Y} \sim \text{MVN}(\mathbf{XB} + \mathbf{ZU}, \boldsymbol{\Sigma}_E)$$

Priors

$$\mathbf{B} \sim \mathbf{U}(-Inf, Inf)$$

$$\mathbf{U} \sim \mathbf{MVN}(\mathbf{0}, \mathbf{\Sigma}_U)$$

$$\mathbf{\Sigma}_U = \mathbf{diag}(\mathbf{S})_U \mathbf{\Omega}_U \mathbf{diag}(\mathbf{S})_U$$

$$\mathbf{S}_U \sim \mathbf{Cauchy}(0, 5)$$

$$\mathbf{\Omega}_U \sim \mathbf{LKJ}(1)$$

brms Animal Model Formula

```
brmf <- brmsformula(  
  mvbind(  
    milk,  
    fat  
  ) ~ 1 +  
  factor(lact) +  
  herd +  
  (1|g|gr(idA, cov = A)) + # additive genetic effect  
  (1|p|gr(id)), # PE effect  
  data = D,  
  data2 = list(A = A)  
)
```

Check Stan Code

```
stanCode <- make_stancode(  
  formula = brmf,  
  data = D,  
  data2 = list(A = A),  
  family = gaussian()  
)
```

```
stanData <- make_standata(  
  formula = brmf,  
  data = D,  
  data2 = list(A = A),  
  family = gaussian()  
)
```

```
stanCode
```

Resources

Stan

brms

<https://github.com/cheuerde/AnimalModels>

<https://github.com/cheuerde/BayesianAlphabet>